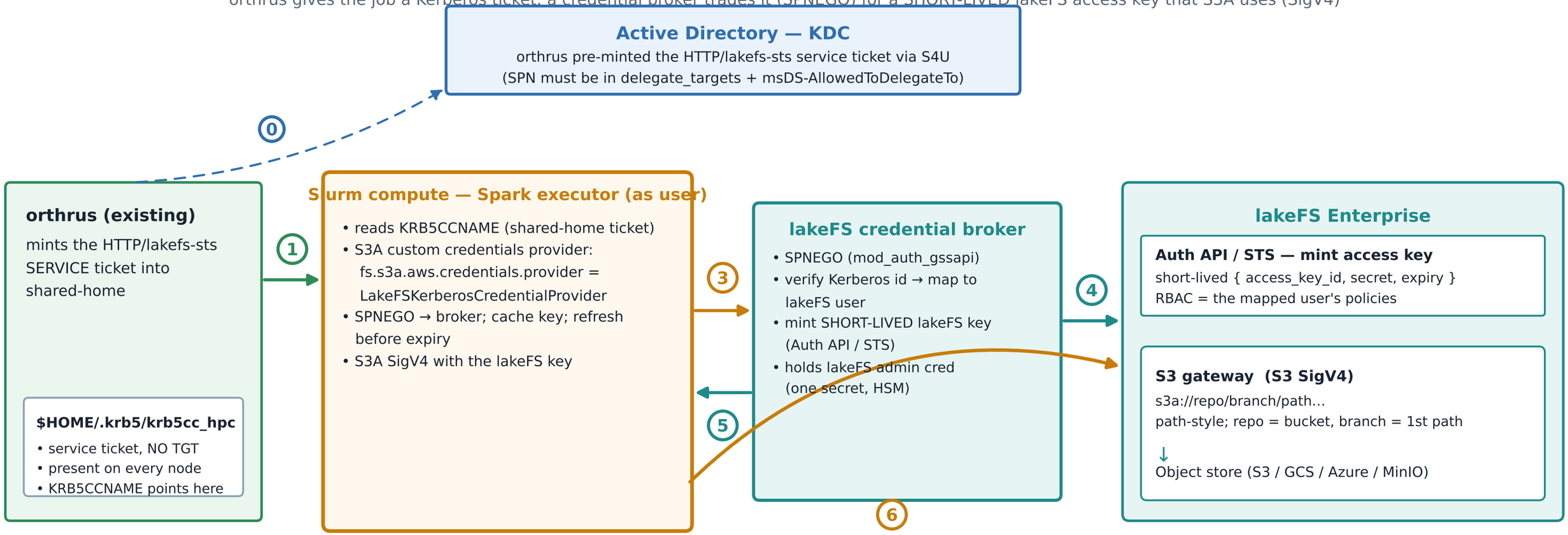
Slurm / Spark → lakeFS Enterprise over S3A, with orthrus-bridged credentials

orthrus gives the job a Kerberos ticket; a credential broker trades it (SPNEGO) for a SHORT-LIVED lakeFS access key that S3A uses (SigV4)



Flow

0. (prereq) the broker SPN HTTP/lakefs-sts is in orthrus's delegate_targets and the broker's msDS-AllowedToDelegateTo.
1. orthrus mints that service ticket into $HOME/.krb5/krb5cc_hpc on every node (no TGT — only pre-minted service tickets).
2. The Spark executor (running as the user) reads KRB5CCNAME; its S3A credentials provider finds the HTTP/lakefs-sts ticket.
3. The provider calls the credential broker over SPNEGO (Authorization: Negotiate) — targeting exactly that SPN (no TGT to fetch another).
4. The broker verifies the Kerberos identity, maps it to a lakeFS user, and mints a SHORT-LIVED lakeFS access key via the Auth API / STS.
5. The broker returns { access_key_id, secret_access_key, expiry }; the provider caches it and refreshes via SPNEGO before expiry.
6. S3A uses the key (S3 SigV4) against the lakeFS S3 gateway: spark.read/write on s3a://repo/branch/… ; RBAC = the mapped user.